



Chronic Obstructive Pulmonary Disease (COPD)

(Chronic Bronchitis; Emphysema)

By **Robert A. Wise**, MD, Johns Hopkins Asthma and Allergy Center

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Chronic obstructive pulmonary disease is persistent narrowing (blocking, or obstruction) of the airways occurring with emphysema, chronic obstructive bronchitis, or both disorders.

- Cigarette smoking is the most important cause of chronic obstructive pulmonary disease.
- People develop a cough and eventually become short of breath.
- Diagnosis is made with chest x-rays and tests of lung function.
- Stopping smoking and taking medications that help keep airways open are important.
- People who have severe disease may need to take other medications, use oxygen, have pulmonary rehabilitation, or rarely lung volume reduction surgery.

In the United States, about 16 million people have chronic obstructive pulmonary disease (COPD). It is a common cause of death, accounting for more than 140,000 deaths each year.

Worldwide, the number of people with COPD is increasing. Factors contributing to COPD include an increase in smoking in many countries and, throughout the world, exposure to toxins in biomass fuels such as wood and grasses. Death rates may be increasing in medically underserved countries.

COPD leads to a persistent decrease in the rate of airflow from the lungs when the person breathes out (exhales), which is called chronic airflow obstruction. COPD includes the diagnoses of chronic obstructive bronchitis and emphysema. Many people have both disorders.

- **Chronic bronchitis** is defined as cough that produces sputum for at least 3 months during 2 successive years. When chronic bronchitis involves airflow obstruction, it qualifies as chronic obstructive bronchitis.
- **Emphysema** is defined as widespread and irreversible destruction of the alveolar walls (the cells that support the air sacs, or alveoli, that make up the lungs) and enlargement of many of the alveoli.

Chronic asthmatic bronchitis is similar to chronic bronchitis. In addition to having a cough that produces sputum, people have **wheezing** and partially reversible airflow obstruction. It occurs predominantly in

people who smoke and have [asthma](#). Some people have both COPD and asthma. People with both disorders are treated primarily for asthma.

The small airways (bronchioles) of the lungs contain smooth muscles and are normally held open by their attachments to alveolar walls. In emphysema, the destruction of alveolar wall attachments results in collapse of the bronchioles when a person exhales, causing airflow obstruction that is permanent and irreversible. In chronic bronchitis, the glands lining the larger airways (bronchi) of the lungs enlarge and increase their secretion of mucus. Inflammation of the bronchioles develops and causes smooth muscles in lung tissue to contract (spasm), further obstructing airflow. Inflammation also causes swelling of the airway passages and secretions in them, further limiting airflow. Eventually, the small airways in the lung become narrowed and destroyed. [Asthma](#) is also characterized by airflow obstruction. However, unlike airflow obstruction in COPD, airflow obstruction in asthma is completely reversible in most people, either spontaneously or with treatment.

Airflow obstruction in COPD causes air to become trapped in the lungs after a full exhalation, increasing the effort required to breathe. Also in COPD, the number of capillaries in the walls of the alveoli decreases. These abnormalities impair the [exchange of oxygen and carbon dioxide](#) between the alveoli and the blood. In the earlier stages of COPD, oxygen levels in the blood may be decreased, but carbon dioxide levels remain normal. In the later stages, carbon dioxide levels increase and oxygen levels fall.

The [COVID-19](#) pandemic posed a particular risk to people with COPD. Having COPD increased a person's risk of being hospitalized or dying from COVID-19.

Causes of COPD

The **most important** cause of COPD is

- Cigarette smoking

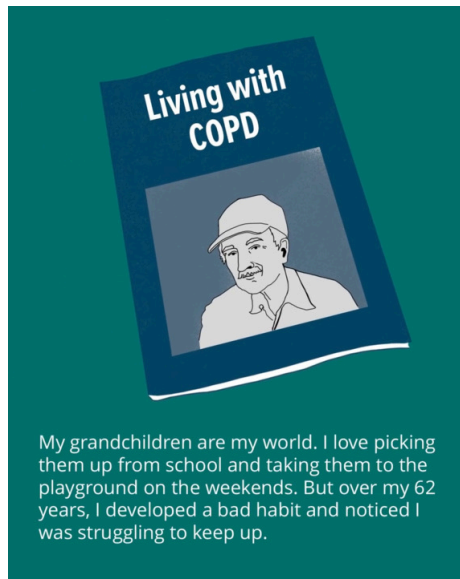
Only about 15% of people who smoke develop COPD. With aging, susceptible people who smoke cigarettes lose lung function more rapidly than people who do not smoke. Lung function improves only a little if people stop smoking. However, the rate of decline of lung function returns to that of people who do not smoke when people stop smoking, thus delaying development and progression of symptoms.

Working in an environment polluted by chemical fumes or dust or by heavy smoke from indoor cooking fires may increase the risk of COPD (see [Overview of Environmental and Occupational Lung Disease](#)). Exposure to air pollution and to smoke from people who are smoking cigarettes nearby (secondhand or passive smoke exposure) may cause flare-ups in people who have COPD but probably does not cause COPD.

COPD tends to occur more often in some families, so there may be an inherited tendency in some people.

Hidden Disability: Chronic Obs...

VIDEO



Low body weight and childhood respiratory disorders contribute to the risk of COPD. However, these factors are of much less importance than cigarette smoking.

A rare cause of COPD is a hereditary condition in which the body does not produce enough of the protein alpha-1 antitrypsin. The main role of this protein is to prevent neutrophil elastase (an enzyme in certain white blood cells) from damaging the alveoli. Consequently, emphysema develops by early middle age in people with severe [alpha-1 antitrypsin deficiency](#) (also called alpha-1 antiprotease deficiency), especially in those who also smoke.

Symptoms of COPD

COPD takes years to develop and progress.

Early COPD

In people with COPD, symptoms develop when people are in their 40s or 50s. They first develop a mild cough that produces clear sputum. The cough and sputum production are usually worse when the person first gets out of bed in the morning. Cough and sputum production can persist throughout the day.

[Shortness of breath](#) may occur during exertion. People often think at first that aging or being in poor physical condition is the cause, and they tend to decrease their physical activity in response. Sometimes, shortness of breath first occurs only when the person has a lung infection (usually [bronchitis](#)), during which time the person coughs more and has an increased amount of sputum. The color of the sputum often changes from clear or white to yellow or green.

Progression of COPD

By the time people with COPD reach their middle to late 60s, especially if they continue smoking, shortness of breath during exertion becomes more troublesome.

[Pneumonia](#) and other lung infections occur more often. Infections may result in severe shortness of breath even when the person is at rest and may require hospitalization. Shortness of breath during activities of daily living, such as toileting, washing, dressing, and sexual activity, may persist after the person has recovered from the lung infection.

About one third of people with severe COPD experience severe weight loss. The cause of weight loss is not clear, and causes may differ among different people. Possible causes include shortness of breath that makes eating difficult and increased levels in the blood of a substance called tumor necrosis factor.

People with COPD may intermittently cough up blood, which is usually due to inflammation of the bronchi, but which always raises the concern of [lung cancer](#).

Morning headaches may occur because breathing decreases during sleep, which causes increased retention of carbon dioxide and decreased levels of oxygen in the blood.

As COPD progresses, some people, especially those who have emphysema, develop unusual breathing patterns. Some people breathe out through pursed lips. Others find it more comfortable to stand over a table with their arms outstretched and their weight on their palms or elbows, a maneuver that improves the function of some of the respiratory muscles.

Over time, many people develop a barrel chest as the size of the lungs increases because of trapped air. Low oxygen levels in the blood can give a blue tint to the skin (cyanosis). Cyanosis may be subtle in dark skin.

Fragile areas in the lungs may rupture, permitting air to leak from the lungs into the pleural space, a condition called [pneumothorax](#). This condition often causes sudden pain and shortness of breath and requires immediate intervention by a doctor to remove the air from the pleural space.

COPD flare-up

A flare-up (or exacerbation) of COPD is a worsening of symptoms, usually cough, shortness of breath, and increased sputum production. Sputum color often changes to yellow or green, and fever and body aches sometimes occur. Shortness of breath may be present when the person is at rest and may be severe enough to require hospitalization. Severe air pollution, common allergens, and viral or bacterial infections may cause flare-ups.

During severe flare-ups, people may develop a life-threatening condition called [acute respiratory failure](#). Among the possible symptoms are severe shortness of breath (a feeling likened to being drowned), severe anxiety, sweating, cyanosis, and confusion.

Complications of COPD

If low oxygen levels are not treated with supplemental oxygen, complications can occur. Low oxygen levels in the blood, if not treated, stimulate the bone marrow to send more red blood cells into the

bloodstream, a condition known as [secondary polycythemia](#) (also called erythrocythemia). The low oxygen levels in the blood also constrict the blood vessels leading from the right side of the heart to the lungs, thereby increasing the pressure in these vessels. As a result of the increased pressure, called [pulmonary hypertension](#), failure of the right side of the heart can occur (called [cor pulmonale](#)). [Swelling](#) of the legs develops in people who have right-sided heart failure.

[High carbon dioxide levels](#) cause the blood to become acidic ([respiratory acidosis](#)). People become drowsy and can go into a coma and die if the problem is not corrected.

People with COPD also have an increased risk of developing [heart rhythm abnormalities](#) (arrhythmias). People with COPD who smoke have a higher risk of developing [lung cancer](#) than do people who do not have COPD but smoke the same amount. People with COPD seem to have an increased risk of developing [osteoporosis](#), [depression](#), [coronary artery disease](#), muscle wasting (atrophy), and [gastroesophageal reflux](#). However, it is not clear whether the risk is increased because of COPD or other factors.

Diagnosis of COPD

- Chest x-rays
- Tests of lung function

Doctors diagnose chronic bronchitis based on the person's history of a prolonged productive cough for at least 3 months during 2 consecutive years. Doctors must exclude other causes of chronic cough before diagnosing chronic bronchitis.

Emphysema is diagnosed on the basis of findings observed during a physical examination and on pulmonary function test results. However, by the time the doctor notices these abnormalities, emphysema is moderately severe. Findings on a chest x-ray or computed tomography (CT) of the chest may also help in diagnosis of emphysema and sometimes chronic bronchitis. It is not important for doctors to differentiate between chronic bronchitis and emphysema, and often chronic bronchitis and emphysema occur together in the same person. The most important determinant of how the person feels and functions is the severity of the airflow obstruction.

In mild COPD, doctors may find nothing unusual during the physical examination. As the disease progresses, doctors may hear wheezing or notice a decrease in the normal sounds of breathing (decreased breath sounds) when they listen to the lungs with a stethoscope. Doctors may note that it takes a long time for the person to exhale air that has been inhaled (prolonged expiration). Chest movement diminishes during breathing, and the person may use the neck and shoulder muscles to breathe.

In mild COPD, the chest x-ray is usually normal. As COPD worsens, the chest x-ray shows that the lungs contain excess air (over-inflation of the lungs). Over-inflation, thinning of blood vessels, or the presence of cysts in the lungs (called bullae) suggests the presence of emphysema.

Tests of lung function

Doctors can evaluate airflow obstruction with forced expiratory spirometry (tests that measure how much and how quickly air can be exhaled from the lungs—see [Pulmonary Function Testing](#)). Decreases in the maximum amount of air a person can exhale in one second (forced expiratory volume in 1 second, or FEV1) and the ratio of the FEV1 to the amount of air that a person can force out of the lungs after taking the deepest breath possible (forced vital capacity, or FVC) are required to demonstrate airflow obstruction and to make the diagnosis.

Doctors may measure the amount of oxygen in the blood by using a sensor placed on a finger or an earlobe (pulse oximetry) or by taking a sample of blood from an artery ([arterial blood gas analysis](#)) or from a vein. Oxygen levels tend to be low in people with COPD. High levels of carbon dioxide in arterial (and venous) blood occurs late in the course of the disease.

In people who develop COPD at a young age, especially when there is a family history of COPD, the level of alpha-1 antitrypsin in the blood is measured to determine whether [alpha-1 antitrypsin deficiency](#) is present. This genetic disorder is also suspected when COPD develops in people who have never smoked.

Other tests

Sometimes doctors also test heart function with [electrocardiography](#) (ECG) or examine the heart with [echocardiography](#) to ensure that a heart disorder is not causing shortness of breath.

Doctors may do other tests to detect other disorders that could be causing the person's symptoms.

Testing in COPD flare-ups

When people have a COPD flare-up, doctors often use arterial blood gas analysis to determine how much oxygen and carbon dioxide are in the person's blood. Doctors may do a chest x-ray to look for evidence of a lung infection. If they suspect a lung infection, particularly if the flare-up is severe enough that the person is being treated in the hospital, doctors will do additional tests to identify the virus or bacterium causing the flare-up because treatment depends on which organism is responsible.

Treatment of COPD

- Smoking cessation
- Relief of symptoms (for example, with medications)
- Supportive care (for example, pulmonary rehabilitation, a nutritious diet)

The most important treatment for COPD is to [stop smoking](#). Stopping smoking when the airflow obstruction is mild or moderate often lessens cough, reduces the amount of sputum, and slows the development of shortness of breath. Stopping smoking at any point in the disease process provides some benefit. Trying several strategies at once is most likely to be effective. Among these strategies are committing to a specific date for quitting, using behavioral modification techniques (for example, making cigarettes difficult to obtain or rewarding oneself for abstaining for increasingly long periods of time), group counseling and support sessions, and nicotine replacement (for example, by chewing

nicotine gum, wearing a nicotine skin patch, or using a nicotine inhaler, nicotine lozenge, or nicotine nasal spray). The drugs varenicline and bupropion may also help decrease tobacco craving. However, even with the most effective methods, less than half of people who try have quit smoking after one year.

People should also try to avoid exposure to other airborne irritants, including secondhand smoke and air pollution.

Contracting influenza or developing pneumonia may worsen COPD markedly. Therefore, all people with COPD should receive an [influenza vaccination](#) every year. [Pneumococcal vaccination](#) as well as vaccination against [COVID-19](#) and [respiratory syncytial virus](#) (RSV) also help.

Because COPD can cause severe weight loss, people should eat a balanced, nutritious diet.

Treatment of symptoms

Wheezing and shortness of breath are relieved when airflow obstruction decreases. Although airflow obstruction due to emphysema is not reversible, bronchial smooth muscle spasm, inflammation, and increased secretions are all potentially reversible.

Inhaled bronchodilators are given with a device that allows the user to spray a specific and consistent dose of a medication into the airways via the mouth and throat (inhalers, including metered-dose inhalers and dry-powder inhalers). Inhaled bronchodilators include

- Anticholinergic medications
- Beta-adrenergic medications

Both anticholinergic and beta-adrenergic medications relax muscles around the bronchioles.

Anticholinergic medications include ipratropium, umeclidinium, aclidinium, revefenacin, and tiotropium. Ipratropium is given about 4 times daily, aclidinium is given twice per day, and tiotropium, revefenacin, and umeclidinium are given once per day.

Inhaled short-acting beta-adrenergic medications, such as albuterol, more rapidly relieve shortness of breath than anticholinergic medications and so can be most useful during flare-ups. Salmeterol, formoterol, arformoterol, vilanterol, olodaterol, and indacaterol are long-acting beta-adrenergic medications. Salmeterol, arformoterol, and formoterol are given every 12 hours. Indacaterol, olodaterol, and vilanterol are given once a day. Umeclidinium and vilanterol are available as an inhaled formulation that combines both drugs. Olodaterol is also available in combination with tiotropium. Long-acting beta-adrenergic medications are useful for prolonged relief of symptoms in some people, especially at night, but they should not be used for rapid relief of symptoms.

Anticholinergic and beta-adrenergic medications are available in combination. Glycopyrrolate is an anticholinergic medication available in combination with the long-acting beta-adrenergic medications formoterol or indacaterol.

Many people can use metered-dose inhalers more effectively when they inhale the drug through a delivery device called a spacer (see figure [How to Use a Metered-Dose Inhaler](#)). Inhaled bronchodilators may also be given using nebulizers. This mode of therapy is usually reserved for people who have severe disease or for those who cannot use a metered-dose inhaler properly. A nebulizer creates a mist of medication, and the timing of its inhalation does not have to be coordinated with breathing. Nebulizers are portable, and some units can even be plugged into the accessory socket in a car.

Corticosteroids are helpful for many people with moderate or severe COPD whose symptoms cannot be controlled by the other medications or for those who get frequent flare-ups despite the use of other medications. Inhaled corticosteroids do not prevent decline of lung function over time. However, their use decreases symptoms and the frequency of COPD flare-ups. Because the medication is delivered to the lungs, typical doses of inhaled corticosteroids cause fewer side effects than corticosteroid treatment given by mouth. However, high doses of inhaled corticosteroids can have effects throughout the body, such as worsening of [osteoporosis](#), particularly in older adults. Corticosteroids given by mouth are largely restricted to treatment of COPD flare-ups or are given to people who have periodic flare-ups or continue to have symptoms due to airflow obstruction and who are not responding to a simpler regimen.

Phosphodiesterase 4 inhibitors, such as roflumilast, reduce inflammation and widen the airways. Phosphodiesterase 4 inhibitors can be used together with other bronchodilators to reduce the risk of COPD flare-ups. Common side effects include nausea, headache, and weight loss, but these effects may decline with continued use of the medications.

Taking the antibiotic azithromycin or erythromycin long term may help prevent flares of COPD, particularly among people prone to frequent or severe flares and who do not smoke.

Theophylline is given infrequently, only to people who do not respond to other medications. The dose must be carefully controlled by the doctor, and, in some people, levels of the medication in the blood must be measured periodically. A long-acting form of the medication helps to control shortness of breath at night.

Doctors used to give medications (expectorants) to help thin secretions and make them easier to cough up. However, there is no good evidence that these medications work. However, avoiding dehydration may prevent thickening of secretions. Generally the goal is to drink enough fluids to keep the urine pale except for that passed first in the morning.

Pulse oximetry is often used to monitor symptoms. Taking a sample of blood from an artery or vein and measuring the amount of oxygen and carbon dioxide in the blood can provide additional information that is useful in monitoring severe disease.

Treatment of flare-ups

Flare-ups should be treated as soon as possible. When bacterial infection is suspected, a 7- to 10-day course of antibiotic treatment is usually prescribed. Many doctors give people who have COPD a supply of an antibiotic to be kept on hand and taken early in a flare-up. Medications may need to be switched in people in whom the original antibiotics did not work, people who have severe symptoms, and people

who are at risk of infection with organisms that are not likely to be eliminated by the usual medications (resistant bacteria). People whose immune system is suppressed or those who live in nursing homes are most likely to be infected with resistant bacteria.

People with severe flare-ups require hospitalization and treatment with short-acting beta-adrenergic medications, short-acting anticholinergic medications (eg, ipratropium), and corticosteroids given by mouth or by vein, and oxygen. They may require [noninvasive positive-pressure ventilation](#) (for example, pressure support or positive airway pressure ventilation by face mask) or ([mechanical ventilation](#) with placement of an endotracheal (breathing) tube.

Some people with severe disease or frequent flare-ups benefit from taking antibiotics long-term. Commonly given antibiotics include azithromycin, clarithromycin, or erythromycin. However, long-term antibiotic use may not be possible or recommended because of troublesome side effects or because long-term use may make bacteria resistant to the antibiotic's effects.

Oxygen therapy

Some people with COPD need to receive extra oxygen to maintain sufficient oxygen in the blood. Some people need oxygen therapy for only a short time, for example when they are released from a hospital after a COPD flare-up. Long-term [oxygen therapy](#) prolongs the life of people who have advanced COPD and severely reduced oxygen levels in their blood. Although round-the-clock therapy is best, using oxygen 12 hours a day also has some benefits. This therapy reduces the excess of red blood cells caused by low blood oxygen levels ([secondary polycythemia](#)) and helps to relieve [cor pulmonale](#) caused by COPD. Oxygen therapy may also decrease shortness of breath during exercise.

Different devices are available for oxygen therapy. Electrically driven oxygen concentrators are used when electrical outlets are available. Compressed oxygen is available in small tanks that permit people to travel outside of their homes for 2 to 6 hours. Liquid oxygen systems are more expensive but are preferable for active people as they permit several hours away from the source reservoir. Battery-driven portable oxygen concentrators are another option and can be used while travelling on commercial airplanes. People must never use oxygen therapy near open flames or while smoking.

Pulmonary rehabilitation

[Pulmonary rehabilitation](#) can help people who have COPD, but it does not improve lung function. These programs can improve independence and quality of life, decrease the frequency and length of hospital stays, and improve the ability to exercise. Programs encompass

- Education about the disease
- Exercise
- Nutritional counseling
- Psychosocial counseling

People are taught about COPD and given information about when to take antibiotics or contact the doctor about flare-ups.

Exercise programs can be carried out in an outpatient treatment facility or at home. Walking (sometimes on a treadmill) is usually used to exercise the legs. Sometimes stationary bicycling and stair climbing are also used. Weight lifting is used for the arms. Often, oxygen is recommended during exercise. As with any exercise program, gains in conditioning are quickly lost if the person stops exercising.

Special techniques are taught for decreasing shortness of breath during activities, such as cooking, engaging in hobbies, and sexual activity.

Pulmonary rehabilitation programs offer advice and suggestions for eating a balanced nutritious diet to help people avoid the severe weight loss that may occur in COPD.

Although pulmonary rehabilitation programs are usually offered in person, virtual programs are also available and may be particularly useful for people who have difficulty accessing in-person programs.

Other treatments

Over-the-counter cough suppressants are usually of no help and are not recommended.

Opioids are sometimes used to relieve severe coughing spells or pain but should be avoided when possible because they can cause drowsiness and constipation, suppress cough (which can cause or worsen infection) and, if used regularly, can cause dependence or addiction.

For people with a severe [alpha-1 antitrypsin deficiency](#), the missing protein can be replaced. The treatment requires weekly [intravenous infusions of the protein](#).

Lung volume reduction surgery can be carried out in people with severe emphysema in the upper portions of their lungs. The goal is to improve exercise ability and quality of life. In this operation, the most severely diseased portions of the lungs are removed, thus permitting the remaining portions of the lungs and the diaphragm to function better. The improvement can last for at least several years. People are required to stop smoking for at least 6 months before surgery. People should undergo an intense rehabilitation program to determine whether overall function can be improved significantly without surgery before undertaking this operation.

In a different type of procedure, valves can be placed inside parts of the airways to block them off (called endobronchial valves). These valves allow trapped air to escape the lungs. Doing so helps only selected people who have COPD. To place these valves, the doctor uses a flexible viewing tube.

[Lung transplantation](#), of either one lung or both lungs, may be used in certain people who are usually younger than 65 years and have severe airflow obstruction. The goal of lung transplantation is to improve quality of life, because survival time is not necessarily increased. Lifelong immunosuppression is required, placing people at risk of infections.

Prognosis and End-of-Life Issues for COPD

COPD itself usually does not cause death or severe symptoms if the person stops smoking at a time when airflow is only mildly obstructed. Continued smoking, however, virtually ensures that symptoms will worsen. With moderate and severe airflow obstruction, the prognosis becomes progressively worse.

People in advanced stages of COPD are likely to need considerable help with medical care and with activities of daily living. They may, for example, arrange to live on a single floor of their house, eat several small meals each day rather than one large one, and avoid wearing shoes that must be tied.

Death may result from [respiratory failure](#), [lung cancer](#), heart disorders (for example, [heart failure](#) or [heart rhythm abnormalities](#)), [pneumonia](#), [pneumothorax](#), or blockage of the arteries leading to the lungs ([pulmonary embolism](#)).

People with end-stage disease who develop flare-ups may need a breathing tube and [mechanical ventilation](#). The duration of mechanical ventilation may be prolonged, and some people remain ventilator-dependent until death. It is important for people to consider with their doctors and loved ones whether or not they wish this kind of supportive therapy and to do so before a flare-up occurs.

An alternative to this supportive therapy is [treatment aimed at comfort](#) (and not at prolonging life). The best way to ensure that the person's wishes regarding prolonged mechanical ventilation are carried out is to prepare an [advance directive](#) and appoint a [health care proxy](#).

More Information

The following English-language resources may be useful. Please note that THE MANUAL is not responsible for the content of these resources.

[COPD Foundation](#): Includes information on COPD diagnosis and treatment and tools to support people with COPD and their caregivers

[American Lung Association: Chronic Obstructive Pulmonary Disease \(COPD\)](#): Including information about COPD risk factors, diagnosis, and treatment and tools for help in quitting smoking and living with COPD



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